🔐 SOC Internship – Day 9 Notes

📌 Topic: Real Incident Case Study (Brute Force Attack)

🎯 Objective

Understand how SOC analysts investigate real-world security incidents step-by-step using log data.

📖 Scenario Summary

- Multiple failed login attempts detected on a server
- After several failures, a **successful login** occurred
- All attempts came from the **same IP address**

📊 Log Analysis

```
Apr 10 10:17:45 Failed password for user1 from 192.168.1.50
Apr 10 10:18:02 Failed password for user1 from 192.168.1.50
Apr 10 10:18:15 Failed password for user1 from 192.168.1.50
Apr 10 10:18:30 Failed password for user1 from 192.168.1.50
Apr 10 10:20:10 Accepted password for user1 from 192.168.1.50
```

🔍 Investigation Steps

1. **Identify repeated failed login attempts**
   → Multiple failed attempts indicate suspicious activity
2. **Check source IP address**
   → Same IP: 192.168.1.50
3. **Analyze login success**
   → Eventually, login was successful
4. **Determine attack type**
   → Pattern matches a brute force attack
5. **Assess impact**
   → Unauthorized access may have occurred

🧠 Analysis Conclusion

- Same IP repeatedly tried passwords
- Multiple failures followed by success
- Clear indication of a **Brute Force Attack**

⚙️ SOC Workflow Mapping

🟢 L1 Analyst (Monitoring)

- Detects unusual login alerts
- Escalates issue to L2

🟡 L2 Analyst (Investigation)

- Analyzes logs
- Confirms brute force attack
- Blocks suspicious IP
- Secures compromised account

🔴 L3 Analyst (Advanced Response)

- Creates detection rules
- Improves security systems
- Prevents future attacks

❓ Key Questions & Answers

1. What type of attack is this?
→ Brute Force Attack

2. Which IP is suspicious?
→ 192.168.1.50

3. Was the attack successful?
→ Yes (login accepted)

4. What action should be taken?
→ Block IP, reset password, secure account

🛡️ Prevention Methods

- Enable **account lockout policy**
- Use **strong passwords**
- Implement **Multi-Factor Authentication (MFA)**
- Monitor logs regularly
- Use **Intrusion Detection Systems (IDS)**
- Limit login attempts

📝 Assignment Guidance

1. Full Incident Analysis

- Describe attack pattern
- Identify attacker behavior
- Explain impact

2. SOC Team Steps

- Detection → Investigation → Response → Prevention

3. Prevention Suggestions

- Mention practical security controls